

ADMISSION NOTE - Patient synth_02 - Acute heart failure exacerbation

PATIENT: Chen, Robert

MRN: 789012

ADMIT DATE: 08/11/2023

ATTENDING: Dr. M. Williams

CHIEF COMPLAINT: Shortness of breath

HISTORY OF PRESENT ILLNESS: Mr. Chen is a 78-year-old male with a known history of HFrEF (last LVEF 35% 6 months ago), HTN, and T2DM who presents with a 3-day history of progressively worsening dyspnea on exertion and at rest. He reports orthopnea requiring 3 pillows to sleep, up from his baseline of 1 pillow. He also notes new bilateral lower extremity swelling and a self-reported 10-pound weight gain over the past week. He denies chest pain, palpitations, fever, or cough. Patient admits to dietary indiscretion over the weekend, having attended a large family gathering with salty foods. He has been taking his home medications as prescribed.

PAST MEDICAL HISTORY:

1. Congestive Heart Failure with Reduced Ejection Fraction (HFrEF), NYHA Class II at baseline.
2. Coronary Artery Disease s/p CABG x3 in 2015.
3. Hypertension.
4. Type 2 Diabetes Mellitus.
5. Hyperlipidemia.
6. Gout, chronic.

PAST SURGICAL HISTORY: CABG (2015), Appendectomy (1960s).

MEDICATIONS ON ADMISSION: Lisinopril 20mg daily, Carvedilol 12.5mg BID, Furosemide 40mg daily, Atorvastatin 40mg daily, Metformin 1000mg BID, Allopurinol 100mg daily.

SOCIAL HISTORY: Retired high school history teacher. Lives at home with his wife, who is his primary caregiver. 40-pack-year smoking history, quit over 20 years ago. Drinks one glass of wine with dinner socially.

PHYSICAL EXAM:

VITALS: T 98.6F, HR 98 bpm, BP 155/92 mmHg, RR 24, SpO2 90% on room air.

GENERAL: Alert and oriented x3, in mild respiratory distress. Using accessory muscles.

CV: Tachycardic but regular rhythm. Distant heart sounds with an S3 gallop. JVD present to the angle of the jaw at 45 degrees.

LUNGS: Bilateral crackles auscultated halfway up the posterior lung fields.

EXT: 2+ pitting edema bilaterally to the mid-shins.

IMAGING/EKG:

CXR: Shows pulmonary vascular congestion, cardiomegaly, and small bilateral pleural effusions.

EKG: Normal sinus rhythm at 98 bpm, LVH by voltage criteria, no acute ST-T wave changes.

ASSESSMENT/PLAN:

78 M with acute decompensated heart failure exacerbation, precipitated by dietary non-compliance.

1. ****Acute on Chronic HFrEF****: Patient is clearly volume overloaded. Admit to telemetry. Start IV Furosemide 80mg IV q12h. Strict I/O monitoring and daily weights.
2. ****Hypertension****: Likely exacerbated by volume overload. Expect improvement with diuresis. Continue home antihypertensives.
3. ****T2DM****: Continue home Metformin. Monitor glucose levels q6h.
4. ****Rule out Hypothyroidism****: As a potential contributor to decompensation given patient's age and presentation, will order a TSH level.
5. ****Gout****: Continue home Allopurinol.
6. ****VTE Prophylaxis****: Start Heparin 5000 units subcutaneously q8h.

PROGRESS NOTE ? Day 1 - Patient synth_02 - Acute heart failure exacerbation

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DATE: 08/12/2023

ATTENDING: Dr. M. Williams

S: Patient reports feeling 'a bit better' this morning. States his breathing is easier and he was able to sleep with two pillows last night. No chest pain or other new complaints.

O:

VS: T 98.4F, HR 88, BP 142/85, RR 20, SpO2 94% on room air.

Weight: 204 lbs (down 6 lbs from admission weight of 210 lbs).

I/O (24h): 3.5L in / 5.5L out. Net negative 2.0L.

PE: JVD has decreased, now visible 4cm above sternal angle. Lung crackles are now confined to the bases. Extremity edema has improved to 1+ at the ankles.

Labs: BMP this AM: Na 136, K 3.8, Cl 101, BUN 28, Cr 1.4 (baseline 1.2). BNP on admission was 2500 pg/mL.

A/P: Acute HFrEF exacerbation, responding appropriately to IV diuresis.

1. **HFrEF**: Good symptomatic and objective improvement. Continue IV Furosemide 80mg BID. Monitor daily BMP and weight.
2. **HTN**: BP improving with diuresis. Continue home Carvedilol and Lisinopril.
3. **Hypokalemia**: K is at lower end of normal. To prevent further depletion with diuresis, will start KCl 20mEq PO BID.
4. **T2DM**: Blood sugars remain well-controlled. Continue Metformin.
5. **Disposition**: Continue current management. If he continues to improve, may be able to transition to PO diuretics tomorrow for discharge planning.

PROGRESS NOTE ? Day 2 - Patient synth_02 - Acute heart failure exacerbation

PATIENT: Chen, Robert

MRN: 789012

DATE: 08/13/2023

ATTENDING: Dr. M. Williams

S: Patient feels 'much improved' and is eager to return home. He ambulated in the hallway with physical therapy this morning without significant dyspnea or oxygen desaturation. Reports baseline breathing at rest.

O:

VS: T 98.5F, HR 76, BP 128/78, RR 18, SpO2 96% on room air.

Weight: 200 lbs (down 4 lbs from yesterday, 10 lbs total).

I/O (24h): 2.0L in / 3.5L out. Net negative 1.5L.

PE: Exam shows euvoemia. No JVD. Lungs are clear to auscultation bilaterally. Trace ankle edema only.

Labs: BMP this AM: Na 137, K 4.1, Cl 102, BUN 24, Cr 1.3. Renal function stable.

A/P: Acute HFrEF exacerbation, now resolved. Patient is at dry weight and clinically stable for discharge.

1. **HFrEF**: Successfully diuresed to euvoemia. Discontinue IV Furosemide. Will start challenge with oral Furosemide 80mg PO this morning. If UOP remains adequate, proceed with discharge.
2. **Medication Reconciliation**: For discharge, will increase maintenance Furosemide dose to 60mg PO daily (from 40mg pre-admission) to prevent re-accumulation of fluid. Other home medications will be continued.
3. **Education**: Case management has provided extensive education on low sodium diet, daily weights, and symptom monitoring. Patient and wife verbalized understanding.
4. **Disposition**: Plan for discharge home this afternoon. Follow-up appointments with PCP and Cardiology to be arranged.

LABORATORY REPORT - Patient synth_02 - Acute heart failure exacerbation

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--- CHEMISTRY PANEL ---

Test | 08/11 (Adm) | 08/12 | 08/13

Sodium (mmol/L) | 136 | 136 | 137
Potassium (mmol/L) | 3.9 | 3.8 | 4.1
Chloride (mmol/L) | 102 | 101 | 102
BUN (mg/dL) | 26 | 28 | 24
Creatinine (mg/dL) | 1.3 | 1.4 | 1.3
Glucose (mg/dL) | 145 | 138 | 130

--- CARDIAC MARKERS ---

Test | Result | Date

BNP (pg/mL) | 2500 | 08/11

--- HEMATOLOGY ---

Test | Result | Date

WBC (k/uL) | 8.5 | 08/11
Hgb (g/dL) | 12.8 | 08/11
Hct (%) | 38.4 | 08/11
Platelets (k/uL) | 210 | 08/11

--- URINALYSIS ---

Test | Result | Date

Urine Microalbumin/Creatinine | PENDING | 08/12

MEDICATION RECORD - Patient synth_02 - Acute heart failure exacerbation

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--- Medications on Admission (08/11/2023) ---

1. Lisinopril 20 mg Tablet, 1 tablet by mouth daily.
2. Carvedilol 12.5 mg Tablet, 1 tablet by mouth twice a day.
3. Furosemide 40 mg Tablet, 1 tablet by mouth daily.
4. Atorvastatin 40 mg Tablet, 1 tablet by mouth every evening.
5. Metformin 1000 mg Tablet, 1 tablet by mouth twice a day.
6. Allopurinol 100 mg Tablet, 1 tablet by mouth daily.

--- Discharge Medications (08/13/2023) ---

1. Lisinopril 20 mg Tablet, 1 tablet by mouth daily. #30
2. Carvedilol 12.5 mg Tablet, 1 tablet by mouth twice a day. #60
3. Furosemide 60 mg Tablet, 1 tablet by mouth daily. #30 (Note: Dose increased)
4. Atorvastatin 40 mg Tablet, 1 tablet by mouth every evening. #30
5. Metformin 1000 mg Tablet, 1 tablet by mouth twice a day. #60

DISCHARGE ADVICE - Patient synth_02 - Acute heart failure exacerbation

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DISCHARGE DATE: 08/13/2023

****Discharge Diagnoses:****

1. Acute on Chronic Systolic Heart Failure (HFrEF)
2. Hypertension
3. Type 2 Diabetes Mellitus
4. Coronary Artery Disease
5. Hypothyroidism

****Condition at Discharge:**** Stable. Patient is euvolemic, breathing comfortably on room air, and ambulating independently. He is alert and oriented, and his vital signs are stable.

****Discharge Instructions:****

1. ****Diet****: Follow a low sodium diet, limited to 2,000 mg per day. Avoid canned soups, processed meats, and frozen dinners.
2. ****Activity****: You may resume your normal activities as tolerated. Avoid heavy lifting or strenuous exertion for one week.
3. ****Weight****: Weigh yourself every morning after urinating but before eating. Record the weight. Call your doctor if you gain more than 3 pounds in one day, or more than 5 pounds in one week.
4. ****Medications****: Take all medications as prescribed on your new medication list. We have increased your water pill (Furosemide) dose.

****Follow-up:****

- Follow up with your Primary Care Provider, Dr. Smith, in 5-7 days.
- Follow up with your Cardiologist, Dr. Jones, in 2-3 weeks.

****Pending Results:****

- Your ****Urine Microalbumin/Creatinine Ratio**** test results are still pending. The results will be sent directly to your primary care doctor, who will discuss them with you at your follow-up appointment.